Hyperframes and the Epistemology of Machine-Gathered Semantics

Coherence, not truth, and where that leads mathematically

Project `thejeb`

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Abstract

Set theoretic estimation (STE) as mechanized in this project does not start from a crisp assertion "$X = v$." It starts from a *possibility set*: a source commits only to "the referent is somewhere in $S \subseteq \Xi$." We call this primitive object a *hyperframe*, and argue it is the right primitive precisely because the alternative — pretending a text (or a labeler reading a text) hands us a single determinate referent — is false to how meaning is actually gathered, by machine or by hand. This note states that epistemological position plainly, brackets two rival pictures of meaning it is *not* chasing (a Platonic universal meaning, a Chomskyan deep structure resolved in the mind), and is honest about where the project's formal guarantees stop and interpretation begins. It then works out the mathematics that dropping "truth as correspondence" and keeping "coherence as criterion" actually forces: lattice theory and Formal Concept Analysis, rough-set indiscernibility, monotone update semantics, a paraconsistent reading of local contradiction, and a genuine metric geometry on readings. Every claim marked as a *theorem* is machine-checked in the project's Lean library and cited by name; everything else is framing, cited to the literature it borrows from.

1 The primitive is the hyperframe, not the crisp frame

A *crisp frame* says a variable X takes a specific value v. That is the textbook picture, and it is the wrong starting point for text. When a document is read — by a person or by a language model standing in for one — what is actually extracted is rarely "$X = v$." It is closer to "on the evidence in this passage, X is one of S," where S may be a singleton, but need not be. The reader does not know, and the text does not always determine, the exact referent; reference is *underdetermined by design*, not by accident of a lazy extractor. This is Quine's inscrutability of reference in a concrete engineering guise: even fully cooperative, fully competent interpretation leaves room for more than one referent consistent with everything actually said [9].

The project's Lean development takes this underdetermination as the object level, not as noise to be denoised away. In `Ste.HyperFrame`, a source contributes a *possibility set* `S i` $\subseteq \Xi$ over a space of candidate referents Ξ, and satisfaction is literally membership:

$$\texttt{sat S a i := a} \in \texttt{S i}.$$

A crisp frame is the special case $|S\,i| = 1$; nothing downstream requires that specialization. Estimation over a corpus is then the question this note is really about: do the possibility sets contributed by many sources cohere, and what does the object built from that coherence look like?

2 What is bracketed, and why

Two rival pictures of "the meaning" are conspicuously absent from this project, and their absence is a decision, not an oversight.

Not a Platonic meaning. One may privately hold that beneath any author's words there is a mind-independent fact of the matter — an ideal Tree, an ideal Justice — that the text either does or does not correctly describe. This note takes no position on that question and needs none. It is bracketed, in the sense Husserl gave that word: the metaphysical question is suspended, not answered, because the object under study is not the world the text is about but the meaning the text *commits to* [7]. Call this a methodological *epoché*: whatever one believes about correspondence to an external reality, it plays no role in what STE checks.

Not a Chomskyan universal deep structure either. A different, non-Platonic universalism is also declined: the idea that every utterance resolves, in a competent mind, to one shared underlying representation across speakers, languages, and eras — a deep structure of which surface texts are mere transformations [2]. That picture chases a single semantics *behind* the corpus. This project chases the opposite: the semantics *of* the corpus, as specified, document by document, by what each document's author actually committed to. Two corpora need not, and generally will not, resolve to the same underlying meaning even for words that look alike on the page, because there is no "underneath" being posited at all.

What is left, once both universalisms are set aside, is what the Cambridge School of intellectual history calls the object of the exercise: *situated authorial commitment* — what a given writer, in a given idiom, at a given moment, was doing with these words, which is recoverable, if at all, only from context: the text, its contemporaries, its conventions [10]. Recovering it is an act of interpretation that only makes sense from inside a horizon of prior understanding brought to the text and revised against it — the Gadamerian hermeneutic circle [5]. Under this scoping, truth-as-correspondence to an external world is simply the wrong target: a corpus of internally coherent falsehoods (a consistent conspiracy theory, a self-contained myth cycle, a novel's invented history) is a perfectly good object of study, and STE should say so — *coherent*, not *true* — without complaint. That is a feature of the scoping, not an embarrassment to be argued around.

3 The extractor is a fallible, situated labeler

Nothing about using a large language model to pull frames out of a document changes this picture; it only makes the mechanism explicit. An LLM extractor stands in for a human labeler doing the same job by hand, and a human labeler is exactly as fallible and exactly as situated as the model: both bring priors, both make judgment calls about what a passage commits its author to, and neither has privileged access to an author's mind.

Concretely: read Spenser's *The Faerie Queene* against Rowling's *Harry Potter* and try to extract a shared frame schema for, say, QUEST or WIZARD. The two texts will produce *materially different* frames — different slot structures, different obligatory fillers, different implicit background assumptions — not because one extraction is more careful than the other, but because the frames genuinely are specified by, and bounded by, each text's own idiom, allegorical machinery, and narrative conventions. There is no neutral, text-independent WIZARD frame that both authors are imperfectly instantiating. The object of study, stated without hedging, is: *context-bounded texts, and the machine (or human) gathering of the semantics as specified by those texts.* Nothing more universal is on offer, and nothing more universal is being claimed.

4 Two levels of coherence, and only one is machine-checked

Once frames are extracted, two different questions can be asked about them, and the project is careful to keep them separate.

Fidelity (frame $\leftrightarrow$ document). Does an extracted possibility set S_i faithfully render what document i actually commits to? This is an act of interpretation. It is not, and in principle cannot be, kernel-checkable: no proof assistant can compare a formal set to a paragraph of English (or Elizabethan English) and certify that the set says what the paragraph means. The extractor, human or machine, is fallible, and an extracted frame *might not* cohere with its document even when every downstream computation is flawless.

Consistency (frame $\leftrightarrow$ frame). Do the extracted frames cohere with one another — is $\bigcap_i S_i \neq \varnothing$, or, more generally, is the corpus feasible on some constraint set? This is exactly STE feasibility, and it *is* machine-checked: it is a computation over sets, and the project's Lean development proves theorems about it.

The honest statement of what the machinery guarantees is therefore a conditional, not an unconditional truth claim:

> *If* the extracted frames faithfully render what their documents commit to, *then* the feasibility verdict STE returns is sound.

The consequent is a theorem. The antecedent is an empirical, auditable, non-formal claim about extraction quality that the project does not and cannot discharge inside the kernel. This is worth quarantining in plain sight, the same way the project already keeps `native_decide` out of its trusted proof base: not because the excluded step is worthless, but because pretending it is kernel-strength would be dishonest about what has actually been checked.

This scoping is productive, not merely modest. Because fidelity and consistency are separated, STE can partially *audit* its own extractor rather than simply trust it. Two independent renderings of the *same* document that STE-conflict are pure extraction noise — a fidelity failure, localized and re-checkable, and clearly distinguishable from genuine cross-*document* authorial disagreement (which is consistency information, not noise). And because frames can be required to carry provenance spans back to the source text, frame$\leftrightarrow$document coherence is made *inspectable* by a human reviewer even though it is not, and will not be, formally provable.

5 Where that leads us, in a math kind of way

Bracketing truth-as-correspondence and adopting coherence-as-criterion is not a retreat into vagueness. It relocates the target of formal work: meaning is no longer analyzed as a proposition that is true or false of the world, so the natural home for it is no longer the logic of propositions. It becomes an object in *order theory* and *closure systems* instead — and that relocation has real, provable content. Five consequences follow, each with a machine-checked piece already in the Lean development.

1. Meaning is a lattice object (Formal Concept Analysis). The STE satisfaction relation `sat S a i := a ∈ S i` is, on the nose, a *formal context* in the sense of Ganter and Wille [6]: hypotheses ($a \in \Xi$) as objects, constraints ($i \in I$) as attributes. `Ste.HyperFrame` proves that the partial feasibility set on a constraint window J is *definitionally* the lower polar of that context (`partialFeasibilitySet_eq_lowerPolar`), so the hyperframes of a corpus — pairs of a feasible-hypothesis set and its matching constraint signature — are exactly the *formal concepts* of the context and form a complete lattice (the abbreviation `HyperFrame` for Mathlib's `Order.Concept`, with its inherited `CompleteLattice` instance). Every partial feasibility set is a concept extent (`isExtent_partialFeasibilitySet`), and every concept's extent recovers the feasible set of its own intent (`hyperFrame_extent_eq_feasibilitySet`). Meaning, on this picture, is not a truth-value assignment; it is a point in a concept lattice, and STE feasibility *is* Formal Concept Analysis.

2. Reference is a rough-set indiscernibility floor. Two candidate referents can be indistinguishable by every constraint a corpus happens to supply, without being the same referent. `Indisc S a b := ∀ i, a ∈ S i ↔ b ∈ S i` captures this: it is an equivalence relation (`indisc_equivalence`), and it is exactly Pawlak's indiscernibility relation for the STE formal context [8]. No subfamily of constraints can ever separate indiscernible hypotheses (`indisc_mem_iff`) — this is the resolution floor of the corpus, below which reference is irreducibly underdetermined, not a defect to be engineered away. Reference, on this picture, is not delivered as a single object; it is a fixed point of constraint accumulation, and the indiscernibility classes are exactly the fibers of the constraint-signature map on the concept lattice (`indisc_iff_upperPolar_singleton_eq`).

3. Content is monotone and dynamic. Because $\bigcap_{i \in K} S_i \subseteq \bigcap_{i \in J} S_i$ whenever $J \subseteq K$, pure accumulation of evidence can only shrink what remains feasible; over a finite hypothesis space this is cardinality-monotone (`ncard_partialFeasibilitySet_antitone`), and it disproves a naive worry about oscillation: if enforcing a larger constraint set somehow yields a *larger* feasible set, that set was not actually larger — some constraint must have been *dropped* (`lt_ncard_imp_not_subset`). Growth in what is deemed feasible therefore always signals retraction, never pure accumulation, which is exactly the shape update semantics gives information states: possibility sets that only ever shrink on genuine new information and require explicit retraction otherwise [12]. Enforcing two constraint sets together is enforcing their union, and feasibility sets compose by intersection under that union (`partialFeasibilitySet_union`) — readings superpose rather than overwrite.

4. Contradiction is local and paraconsistent, not explosive. A possibility set $S_i(v)$ for a variable v can be a singleton (determinate), empty (over-determined — a local contradiction), or have more than one element (under-determined). This is exactly Belnap's four-valued generalization of classical truth [1], transported from the two-element set $\{\text{true}, \text{false}\}$ to an arbitrary reference space Ξ: none, one, or many admissible values, instead of just true or false. Crucially, emptiness on one variable does not trivialize the rest of the corpus — there is no principle of explosion here, in the spirit of da Costa's paraconsistent logics [3]. When the *global* intersection across all sources is empty, the right object to hand back is not silence but the concept lattice of maximal jointly-feasible sub-readings — an admissible-precisification move in the sense of van Fraassen's supervaluationism [11]: keep every way of resolving the conflict that is itself internally coherent, and let disagreement show up as multiplicity in the lattice rather than as global collapse.

5. Content has geometry. The lattice of readings is not just ordered; it carries a genuine metric. `Ste.PartitionRank` equips the coreference partition lattice with a combinatorial, entropy-free rank $r(P) = |\alpha| - \text{blocks}(P)$ and proves it *submodular* (`rank_submodular`), the counting shadow of Shannon's 1953 observation that entropy is a submodular rank function on the lattice of partitions [4]; this rank is monotone in the corpus (`mustSetoid_eq_sInf`, `rank_mustSetoid_mono`), so more documents can only add forced identifications, never remove them. `Ste.InfoDistance` turns this rank into an honest metric, the counting shadow of Shannon's information distance: symmetric, vanishing exactly on equal readings (`shannonDist_eq_zero_iff`), and satisfying the triangle inequality (`shannonDist_triangle`) — so "how far apart are two readings" is a well-posed question with a real answer, not a metaphor. On top of that metric, `Ste.CouplingRank` makes "genuine cross-topic coupling" measurable rather than impressionistic: the excess rank a partition carries once forced through another (`couplingExcess`) is provably positive exactly when the partitions are not already aligned (`couplingExcess_pos_of_not_le`). And where the corpus really is uncoupled — rectangular, in the project's terms — feasibility becomes a purely local check with no obstruction to gluing (`Ste.HellyConsistency`'s `frame_hasHelly_two`: pairwise consistency of authors already certifies whole-corpus consistency, a frame-theoretic Helly number of 2; `Ste.CechCover`'s `rectangular_cechVanishesCover` and `Ste.CechObstruction`'s `diagonal_cechObstruction` for the discriminating example where that vanishing genuinely fails). The correctness criterion throughout is *internal* coherence — a consistency and closure condition on a lattice — never external correspondence to a fact of the matter outside the corpus.

6 The through-line

None of this is a way of avoiding hard questions about meaning by retreating into formalism. It is the opposite move: bracketing truth-as-correspondence is what *lets* the hard questions be asked in a setting that has theorems — lattices, closure systems, rough-set indiscernibility, paraconsistent possibility spaces, and a genuine metric geometry on readings — instead of a setting (propositional truth of an authorial intention no one can access) that mostly has aphorisms. Section 4's conditional is the hinge the whole argument turns on: STE never claims a document's frame is correct, only that *if* extraction was faithful, feasibility is sound, and the machinery is built so that claim can be inspected, partially audited, and never quietly upgraded to more than it is. Where the fidelity question stops — at the boundary of what a possibility set can say about a paragraph of Spenser or a paragraph of Rowling — is exactly where situated, competent, fallible interpretation has to take over, the way it always has.

References

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